Tao Tao

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Education

University of Maryland, College Park — *Ph.D. in Physics, Major: Machine Learning* *Expected Dec. 2027*
Nanjing University — *B.S. in Physics* *June 2020*

Publications

- **Knowledge Capacity Beyond Independent Facts: Shared Structure, Abstraction and Frequency** **In review**
  ***Tao Tao**, Maissam Barkeshli*
- **Learning Pseudorandom Numbers with Transformers: Permuted Congruential Generators, Curricula, and Interpretability** **ICLR 2026**
  ***Tao Tao**, Maissam Barkeshli* *arXiv:2510.26792*
- **(How) Can Transformers Predict Pseudo-Random Numbers?** **ICML 2025**
  ***Tao Tao**, Darshil Doshi, Dayal Singh Kalra, Tianyu He, Maissam Barkeshli* *arXiv:2502.10390*
- **Learning Permuted Congruential Sequences with Transformers** **NeurIPS 2025 MATH-AI Workshop**
  ***Tao Tao**, Maissam Barkeshli*
- **Also:** Quantum diamond microscope for narrowband magnetic imaging (Phys. Rev. Applied **22**, 054050, 2024); Three-dimensional entanglement on a silicon chip (npj Quantum Information, 2020).

ML Research Experience

Knowledge Distillation in Transformers Feb. 2026 – Present
Advisor: Maissam Barkeshli *University of Maryland*

- Investigating whether distillation lets a small model store *more knowledge per parameter* than standard training: students trained on a larger teacher's soft targets vs. the 2–4 bits/param supervised limit, via our latent-knowledge measure.

Knowledge Capacity of Transformers Beyond Independent Facts Dec. 2025 – Present
Advisor: Maissam Barkeshli *University of Maryland*

- Proposed a measure of *latent* knowledge, crediting what a model learns about an underlying fact rather than any sentence expressing it, extending capacity scaling beyond independent random facts.
- Pretrained hundreds of GPT-2 models (up to 100M params) from scratch on synthetic grammar data, tuning learning rate and weight decay under μP, and fit scaling laws of stored knowledge vs. model size and training budget.
- Found that each parameter stores about **3 bits** of structured data but only **0.5 bits** of the most abstract knowledge, and that frequent facts crowd out rare ones, explained with an associative-memory model.

In-Context Learning of Pseudo-Random Number Generators May 2024 – Dec. 2025
Advisor: Maissam Barkeshli *University of Maryland*

- Showed Transformers *in-context learn* hidden recurrences of linear (LCG, **ICML 2025**) and permuted congruential generators (PCG, **ICLR 2026**), generalizing to unseen parameters, beyond classical attacks that assume a known multiplier and modulus.
- Trained models up to 50M params on 5B tokens (moduli to 2^{22}); depth/width/data sweeps gave scaling laws for the context needed ($m^{1/4}$ for LCGs, $\sqrt{m}$ for PCGs).
- Diagnosed loss plateaus at large moduli and designed a curriculum (smaller-modulus pretraining + decaying data mixing) that removed stagnation, widened the stable LR range, and lifted accuracy from 4% to 97% at $m = 2^{20}$.
- Reverse-engineered the learned algorithms: prime-factor residue embeddings and 2^k-lookback attention heads for LCGs; rotation-invariant bit-pattern embedding clusters for PCGs, shared across moduli, explaining curriculum transfer.

Quantum Sensing with NV Centers May 2022 – May 2024
Advisor: Ronald Walsworth *University of Maryland*

- Built Python simulation and parallel data-analysis pipelines for quantum sensor experiments (Phys. Rev. Applied).

Work Experience

Software Development Intern Oct. 2020 – Jan. 2021
Baidu, Inc. *Beijing, China*

- Co-developed the Python layer compiling quantum machine learning algorithms to superconducting hardware, including pulse-generation algorithms (logical gates to control waveforms) and the circuit simulator used as the classical backend.

Service & Skills

Reviewer: ICLR 2026, NeurIPS 2026

Skills: Python, C++, PyTorch, JAX, Hugging Face; LLM pretraining, distributed training, fine-tuning/LoRA, distillation